

Lymphoid Histology

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All images sourced from Digital Histology

Diffuse Lymphatic Tissue



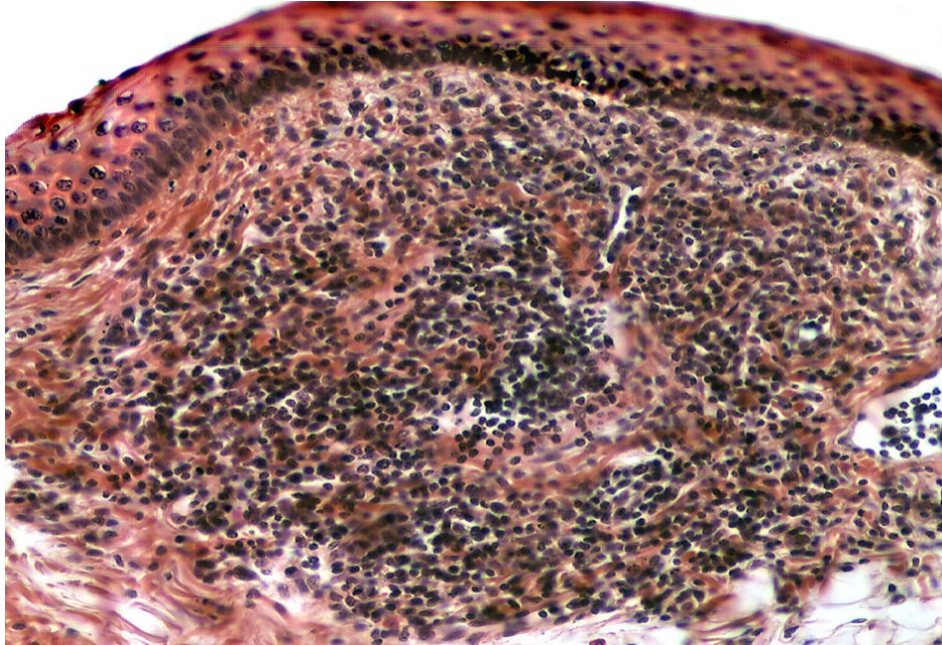
Can be solitary or part of larger lymphoid tissue (tonsil, Peyer's patch)

Made up of scattered **lymphocytes** (dark staining nuclei with scant cytoplasm)

Lymphatic Nodules

More organized cells than diffuse tissue

Primary nodule

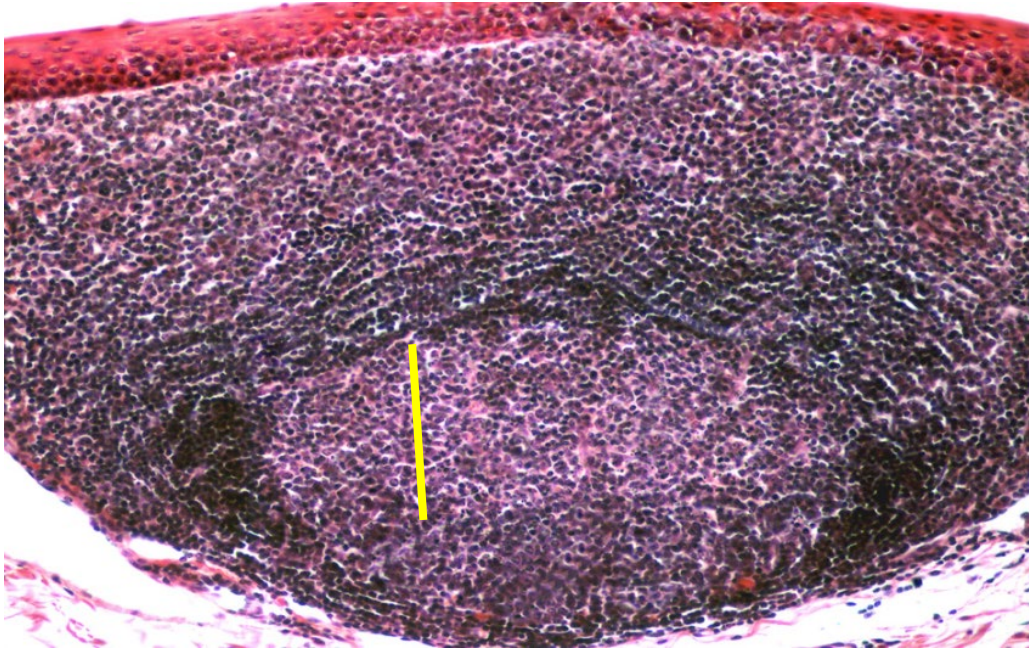


Lymphocytes (dark staining cells) surrounded/supported by **reticular connective tissue**

Reticular connective tissue is a loose connective tissue characterized by a delicate, branching network of reticular fibers (Type III collagen) that form a supportive framework, or stroma, for soft organs like the liver, spleen, lymph nodes, and bone marrow, housing other cells and supporting blood vessels and channels. Produced by specialized fibroblasts called reticular cells, this mesh-like structure provides support, filters blood, and allows immune cells to move freely

Lymphatic Nodules

Secondary nodule



Similar to primary, but
contains a pale
germinal center
(maturing lymphocytes)

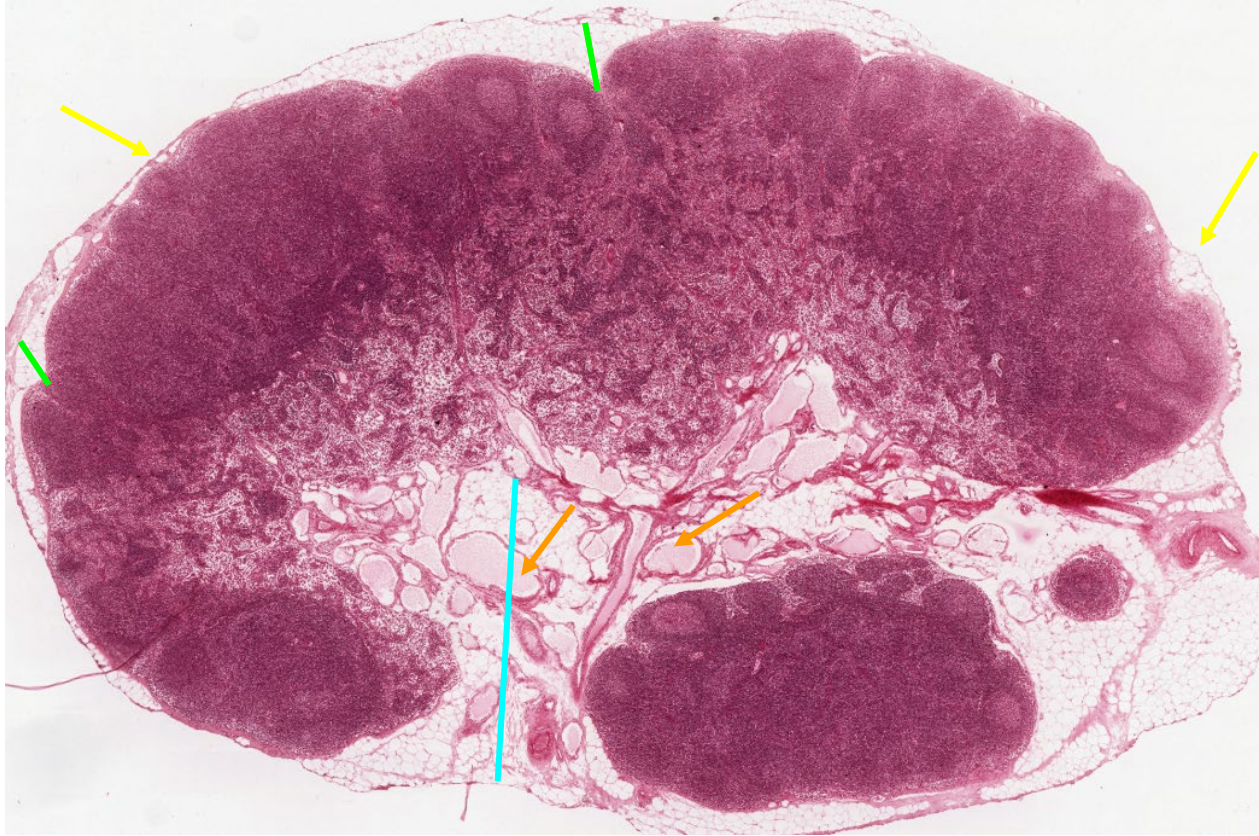
Peyer's Patch



Lymphoid tissue in the mucosa of the ileum (distal small intestine)

Made up of lymphoid nodules and diffuse lymphoid tissue

Lymph Node - Outer Structure



Capsule - dense connective tissue overlying entire node

Subcapsular sinus - space directly under capsule where **afferent lymphatics/sinuses** drain into

Hilum - Indentation where **efferent lymphatics/sinuses** exit node

Lymph Node - Cortex



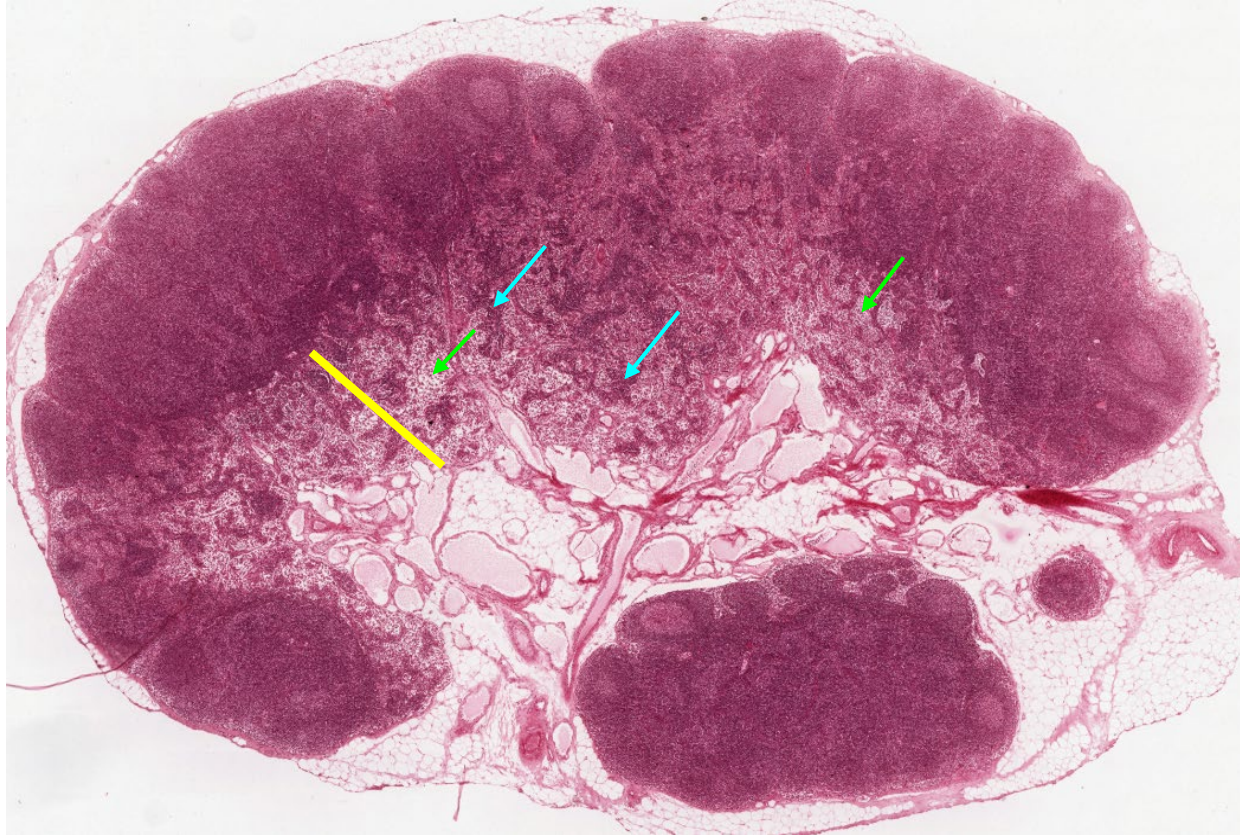
Outer Cortex

- Many lymphoid nodules of B cells

Deep Cortex (aka Paracortex)

- More diffuse tissue with T cells

Lymph Node - Medulla



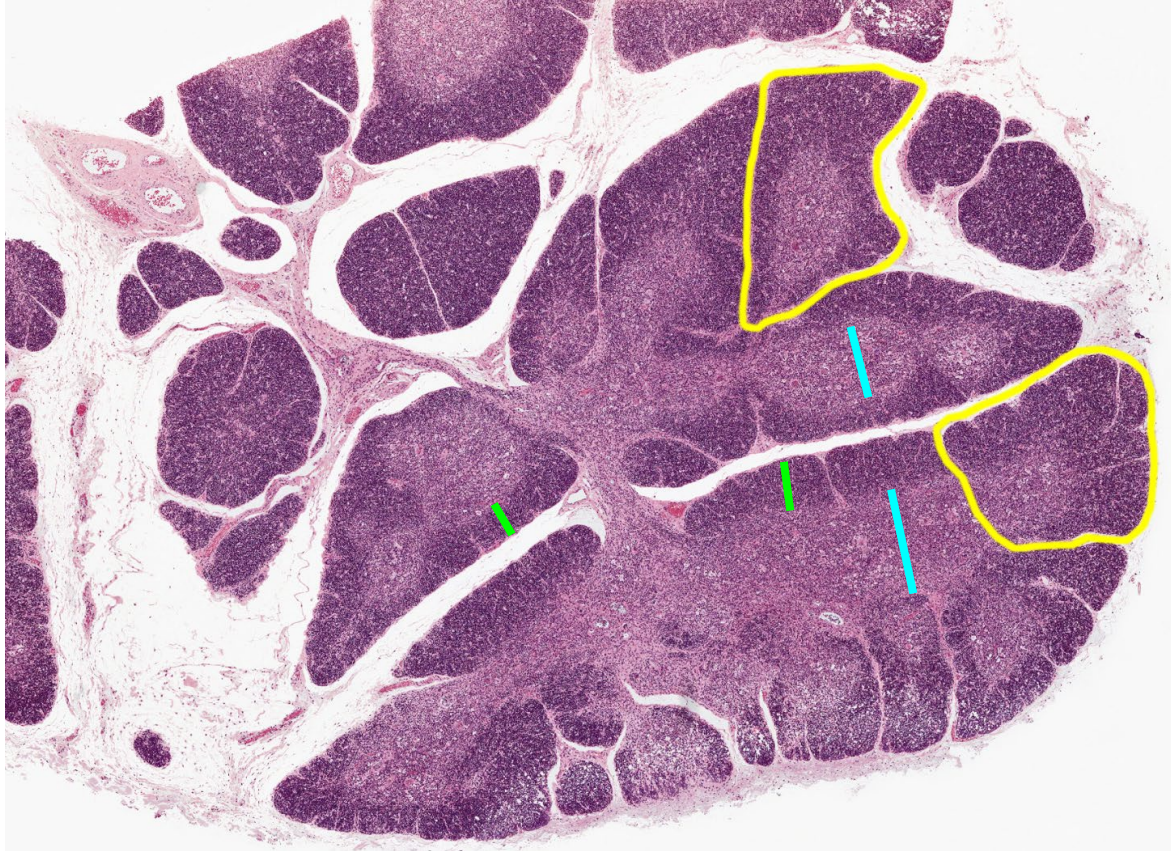
Medullary Cords

- Dark staining projections of B cells

Medullary Sinuses

- Light staining channels where lymph flows

Thymus



Lymphoid organ with distinct **lobules** and 2 layers

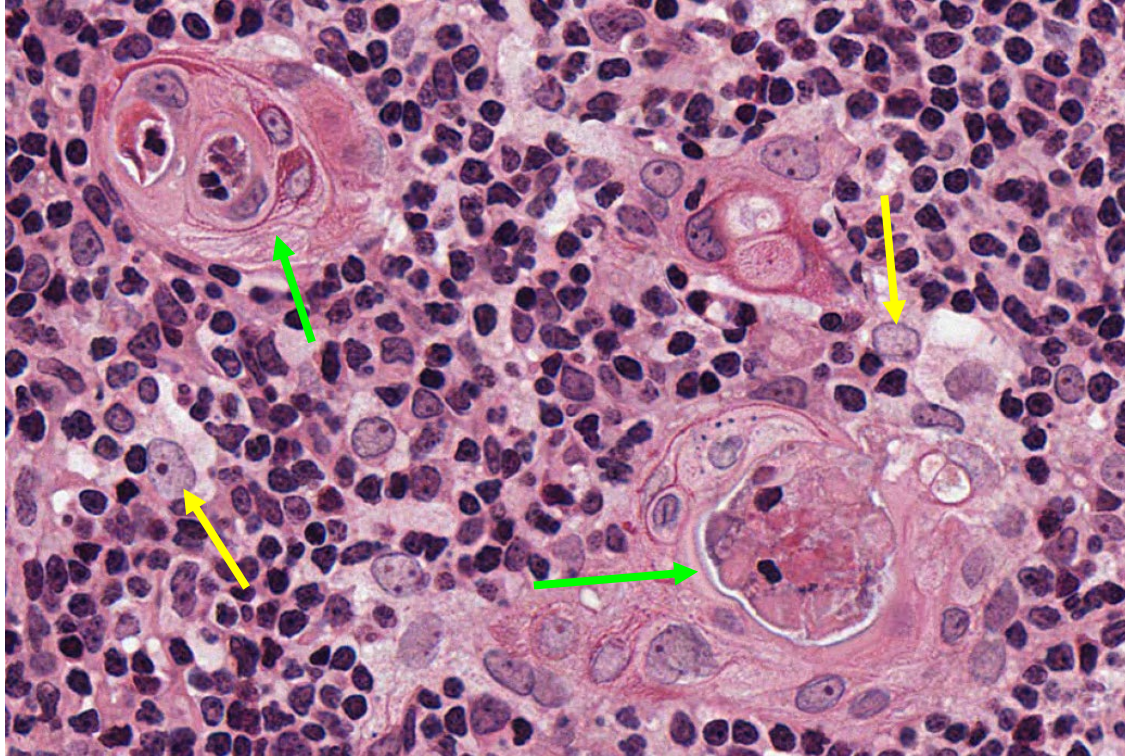
Cortex

- Darker staining, contains immature T cells

Medulla

- Lighter staining, contains more mature T cells

Thymus - Specialized cells



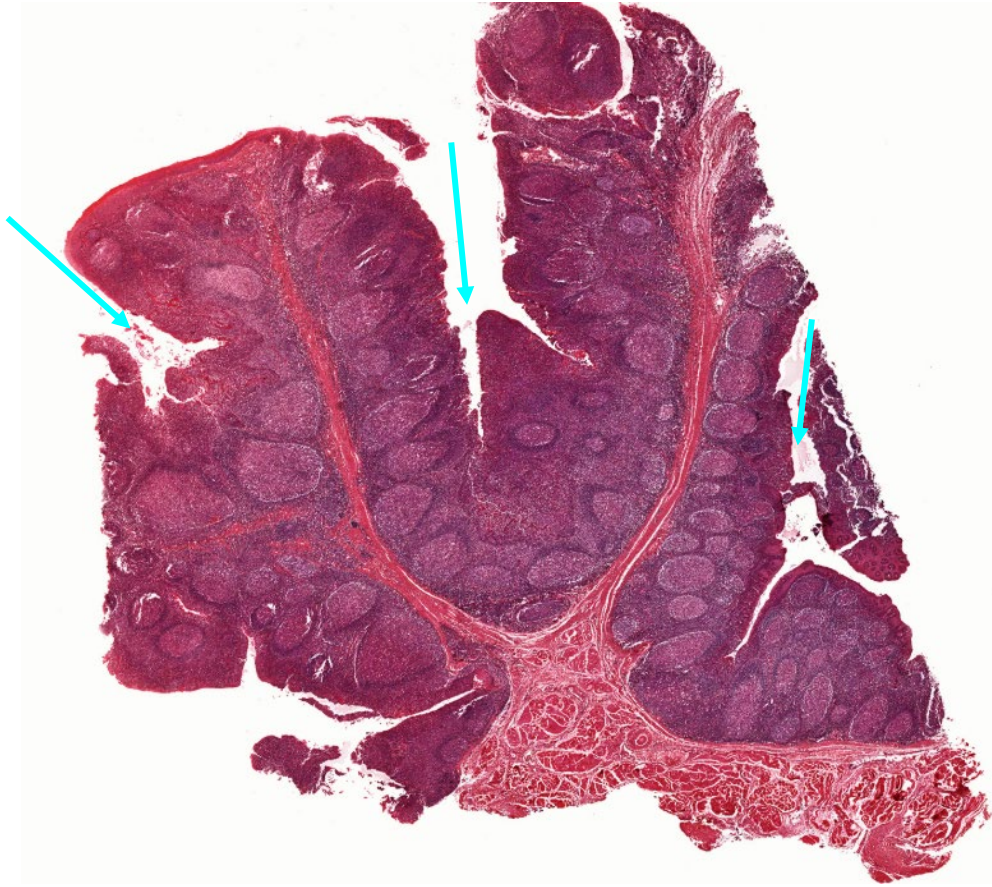
Epithelial reticular cells

- large flat cells of the stroma around lymphoid cells

Hassall's (Thymic) Corpuscles

- Large keratinized structures in the medulla surrounded by ERCs

Tonsil



Many distinct lymphoid nodules

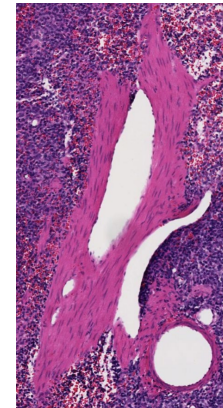
Deep **tonsillar crypts** of
invaginated epithelial tissue

Spleen - Overall Structure

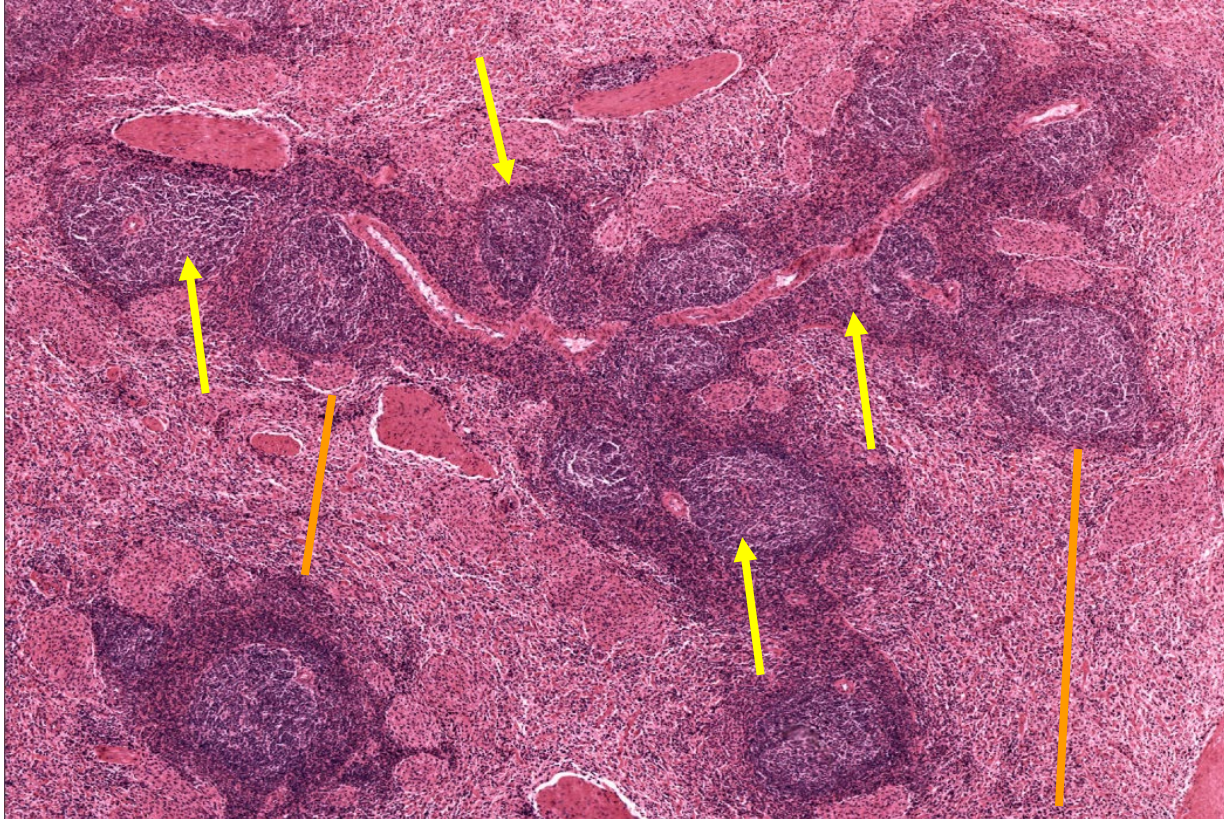


Thick **capsule** with **trabeculae** extending into organ

- Contain branches of splenic artery known as **trabecular arteries**



Spleen - Overall Structure



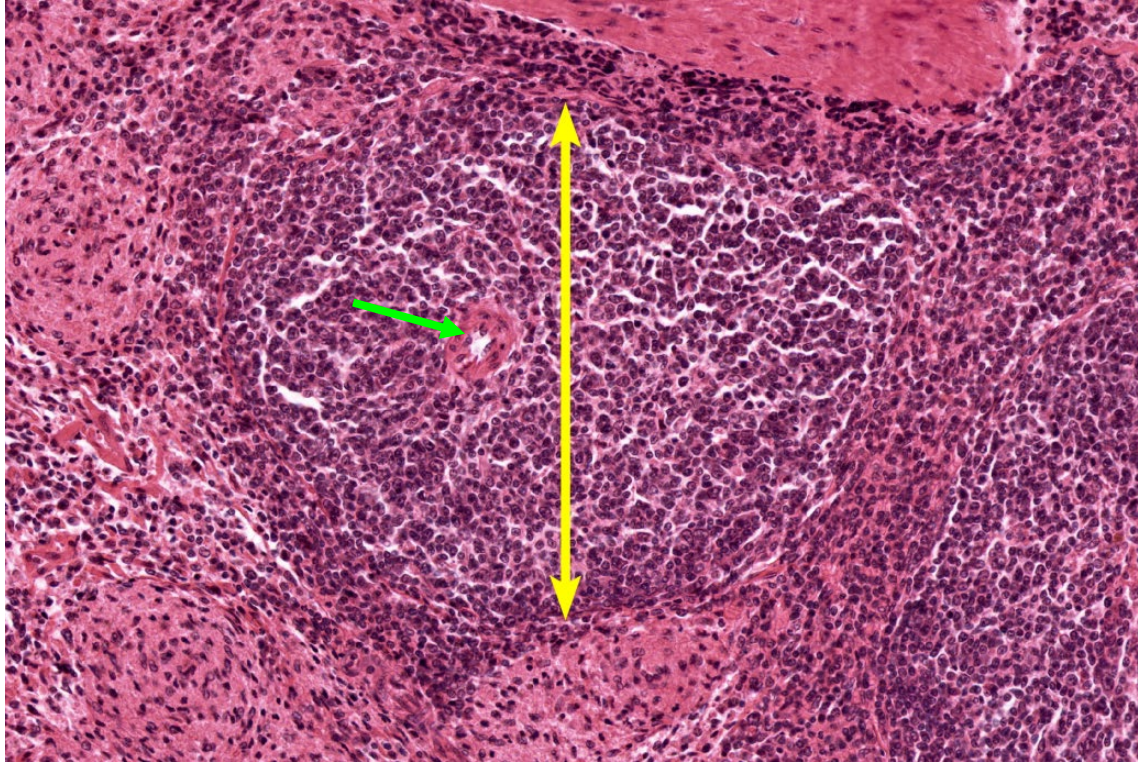
Red Pulp

- Less dense, lighter staining
- Contains splenic sinuses and surrounding cords

White Pulp

- More dense, darker staining
- Contains diffuse and nodular lymphoid tissue

Spleen - White Pulp



Central Arteriole

- Branch of trabecular arteriole
- Surrounded by

Periarterial

Lymphatic Sheath

(PALS)

- Diffuse lymphatic tissue